CONTROL SYSTEM
ETEL 307

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Lecture 1

UNIT I : Control Systems - Basics & Components

1. Introduction to basic terms, classifications & types of Control Systems.
2. Block diagrams & signal flow graphs.
3. Transfer function, determination of transfer function using block diagram reduction techniques and Mason’s Gain formula.
4. Control system components: Electrical/Mechanical/Electronic/A.C./ D.C. Servo Motors, Stepper Motors, Tacho Generators, Synchros, Magnetic Amplifiers, Servo Amplifiers,

Introduction to Control System:

A control system is a system, which provides the desired response by controlling the output. The showed figure shows the simple block diagram of a control system. (Single Block Diagram)

The output is controlled by varying input, We will vary this input with some mechanism.

Examples − Traffic lights control system, washing machine

Classification of Control Systems

Based on some parameters, we can classify the control systems into the following ways.

1. Continuous time and Discrete-time Control Systems

Control Systems can be classified as continuous time control systems and discrete time control systems based on the **type of the signal** used.

- In **continuous time** control systems, all the signals are continuous in time. But, in **discrete time** control systems, there exists one or more discrete time signals.

2. SISO and MIMO Control Systems

Control Systems can be classified as SISO control systems and MIMO control systems based on the **number of inputs and outputs** present.

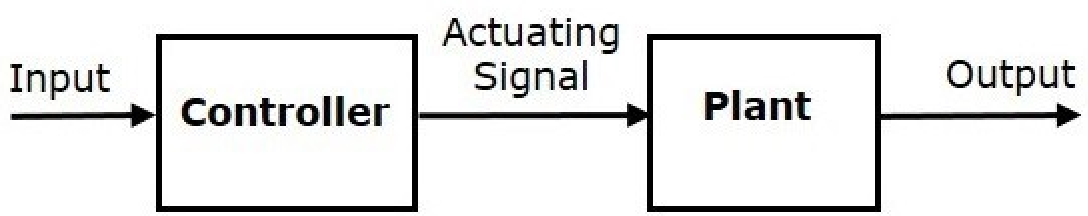
- **SISO** (Single Input and Single Output) control systems have one input and one output. Whereas, **MIMO** (Multiple Inputs and Multiple Outputs) control systems have more than one input and more than one output.

3. Open Loop and Closed Loop Control Systems

Control Systems can be classified as open loop control systems and closed loop control systems based on the **feedback path**.

- In **open loop control systems**, output is not fed-back to the input. So, the control action is independent of the desired output.

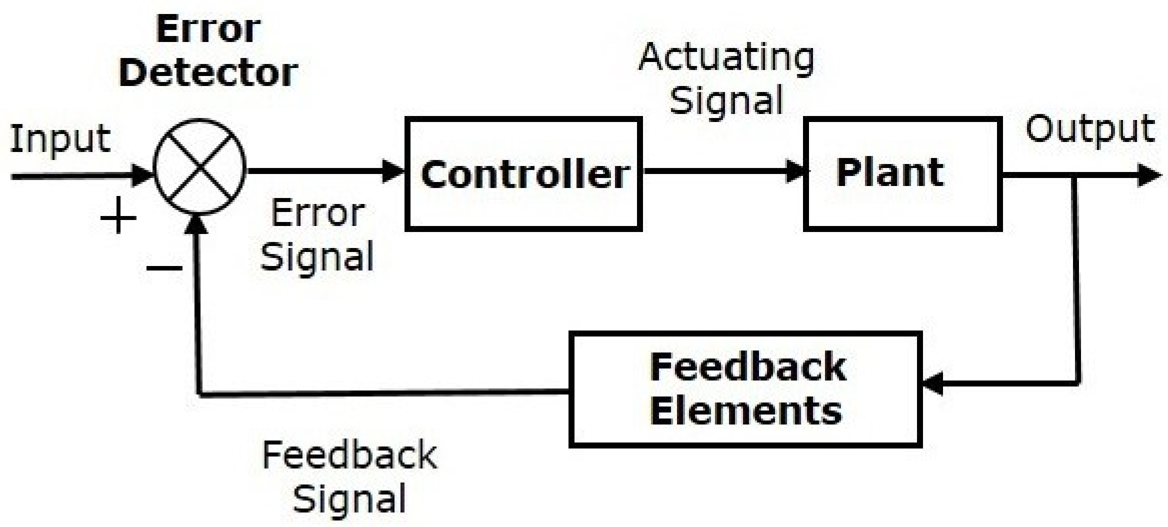
The showed figure shows the block diagram of the open loop control system.



Here, an input is applied to a controller and it produces an actuating signal or controlling signal. This signal is given as an input to a plant or process which is to be controlled. So, the plant produces an output, which is controlled.

- In **closed loop control systems**, output is fed back to the input. So, the control action is dependent on the desired output.

The showed figure shows the block diagram of negative feedback closed loop control system.



Note: The feedback can be positive as well as negative it will be decided according to the system demand.

The error detector produces an error signal, which is the difference between the input and the feedback signal. This feedback signal is obtained from the block (feedback elements) by considering the output of the overall system as an input to this block. Instead of the direct input, the error signal is applied as an input to a controller.

So, the controller produces an actuating signal which controls the plant. In this combination, the output of the control system is adjusted automatically till we get the desired response. Hence, the closed loop control systems are also called the automatic control systems. Traffic lights control system having sensor at the input is an example of a closed loop control system.

The differences between the open loop and the closed loop control systems are mentioned in the following table.

Open Loop Control Systems	Closed Loop Control Systems
Control action is independent of the desired output.	Control action is dependent of the desired output.
Feedback path is not present.	Feedback path is present.
These are also called as **non-feedback control systems**.	These are also called as **feedback control systems**.
Easy to design.	Difficult to design.
These are economical.	These are costlier.
Inaccurate.	Accurate.

FEEDBACK:

If either the output or some part of the output is returned to the input side and utilized as part of the system input, then it is known as **feedback**. Feedback plays an important role in order to improve the performance of the control systems.

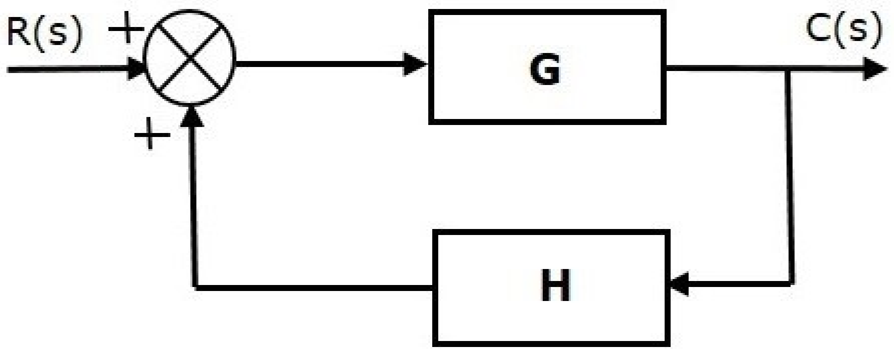
- Types of Feedback

There are two types of feedback

1. Positive feedback
2. Negative feedback

Positive Feedback:

The Positive feedback adds the reference input, R(s) and feedback output. The showed figure is a block diagram of positive feedback control system.



The concept of transfer function will be discussed in later chapters. For the time being, consider the transfer function of positive feedback control system is,

$$T=\frac{G}{1-GH}$$

Where,

- **T** is the transfer function or overall gain of positive feedback control system.
- **G** is the open loop gain, which is function of frequency.
- **H** is the gain of feedback path, which is function of frequency.

Negative Feedback:

Negative feedback reduces the error between the reference input, R(s) and system output. The showed figure is block diagram of negative feedback control system.

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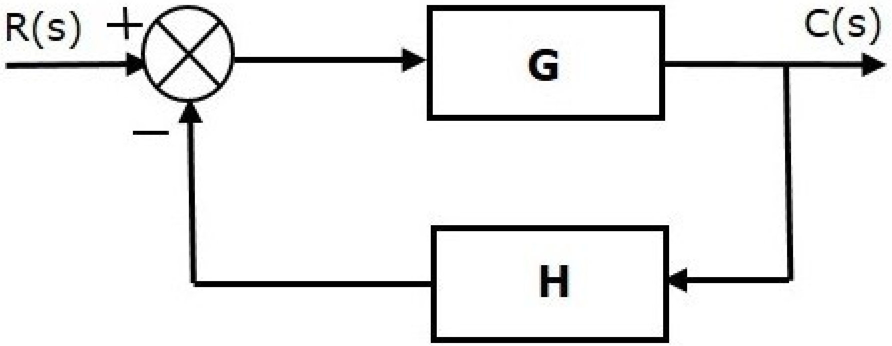

Transfer function of negative feedback control system is,

Transfer function of negative feedback control system is,

$$T = \frac{G}{1+GH}$$

Note: The derivation of above transfer function will be presented in future lecture.

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Effect of Feedback:

1. Effect of Feedback on Overall Gain

the overall gain of negative feedback closed loop control system is the ratio of 'G' and (1+GH). So, the overall gain may increase or decrease depending on the value of (1+GH).

- If the value of (1+GH) is less than 1, then the overall gain increases. In this case, 'GH' value is negative because the gain of the feedback path is negative.
- If the value of (1+GH) is greater than 1, then the overall gain decreases. In this case, 'GH' value is positive because the gain of the feedback path is positive.

In general, 'G' and 'H' are functions of frequency. So, the feedback will increase the overall gain of the system in one frequency range and decrease in the other frequency range.

2. Effect of Feedback on Sensitivity

sensitivity of the overall gain of closed loop control system as the reciprocal of (1+GH). So, Sensitivity may increase or decrease depending on the value of (1+GH).

- If the value of (1+GH) is less than 1, then sensitivity increases. In this case, 'GH' value is negative because the gain of feedback path is negative.
- If the value of (1+GH) is greater than 1, then sensitivity decreases. In this case, 'GH' value is positive because the gain of feedback path is positive.

In general, 'G' and 'H' are functions of frequency. So, feedback will increase the sensitivity of the system gain in one frequency range and decrease in the other frequency range. Therefore, we have to choose the values of 'GH' in such a way that the system is insensitive or less sensitive to parameter variations.

3. Effect of Feedback on Stability

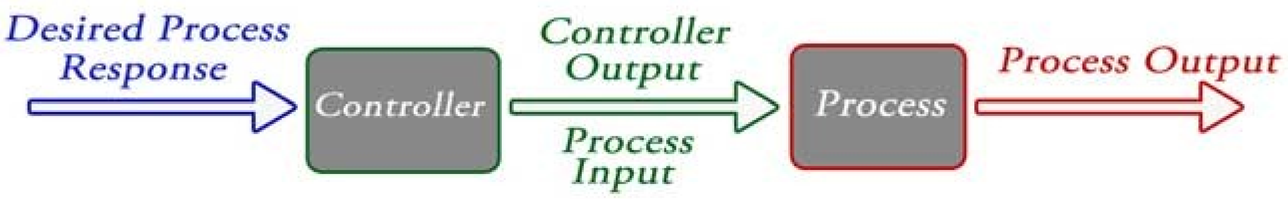
A system is said to be stable, if its output is under control. Otherwise, it is said to be unstable.

- In Transfer function Equation , if the denominator value is zero (i.e., $GH = -1$), then the output of the control system will be infinite. So, the control system becomes unstable.

Therefore, we have to properly choose the feedback in order to make the control system stable.

EXAMPLES OF CONTROL SYSTEM

Open Loop Control System

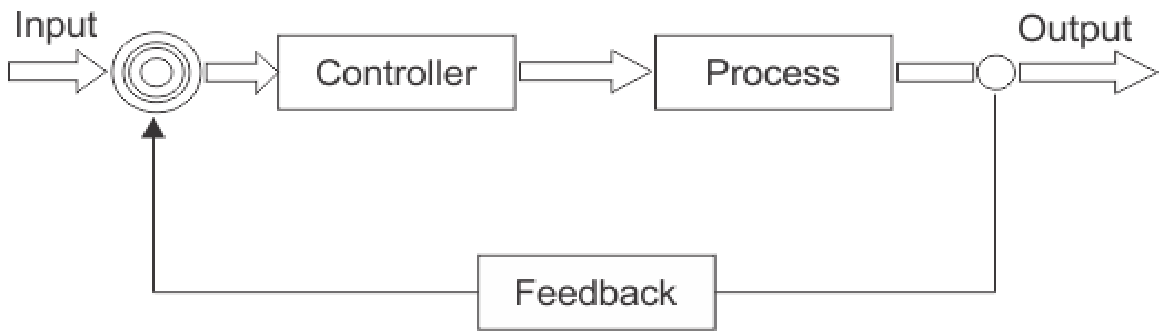


Practical Examples of Open Loop Control System

- Electric Hand Drier – Hot air (output) comes out as long as you keep your hand under the machine, irrespective of how much your hand is dried.
- Automatic Washing Machine – This machine runs according to the pre-set time irrespective of washing is completed or not.
- Bread Toaster – This machine runs as per adjusted time irrespective of toasting is completed or not.
- Automatic Tea/Coffee Maker – These machines also function for pre adjusted time only.

- Timer Based Clothes Drier – This machine dries wet clothes for pre-adjusted time, it does not matter how much the clothes are dried.
- Light Switch – Lamps glow whenever light switch is on irrespective of light is required or not.
- Volume on Stereo System – Volume is adjusted manually irrespective of output volume level.

Closed Loop Control System



Practical Examples of Closed Loop Control System

- Automatic Electric Iron – Heating elements are controlled by output temperature of the iron.
- Servo Voltage Stabilizer – Voltage controller operates depending upon output voltage of the system.
- Water Level Controller – Input water is controlled by water level of the reservoir.
- Missile Launched and Auto Tracked by Radar – The direction of missile is controlled by comparing the target and position of the missile.

- An Air Conditioner – An air conditioner functions depending upon the temperature of the room.
- Cooling System in Car – It operates depending upon the temperature which it controls.

DOMAIN:

1. Time Domain: Time domain refers to variation of amplitude of signal with time. For example consider a typical Electro cardiogram (ECG). If the doctor maps the heartbeat with time say the recording is done for 20 minutes, we call it a time domain signal.
2. Frequency Domain: The frequency domain refers to the analysis of mathematical functions or signals with respect to frequency.

Example: ECG a number of peaks are there (of different types). Say in one heartbeat 4 types of peaks or variation in amplitude occurs. So in frequency domain, over the entire time period of recording, how many times each peak comes is recorded.

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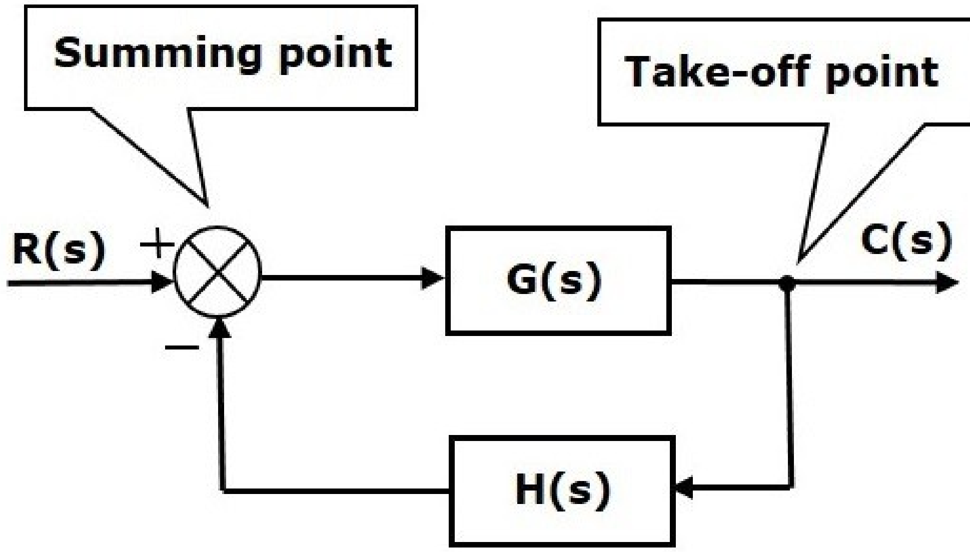
Lecture 2

BLOCK DIAGRAM

Block diagrams consist of a single block or a combination of blocks. These are used to represent the control systems in pictorial form.

Basic Elements of Block Diagram

The basic elements of a block diagram are a block, the summing point and the take-off point. Let us consider the block diagram of a closed loop control system as shown in the following figure to identify these elements.



The above block diagram consists of two blocks having transfer functions G(s) and H(s). It is also having one summing point and one take-off point. Arrows indicate the direction of the flow of signals. Let us now discuss these elements one by one.

Block

The transfer function of a component is represented by a block. Block has single input and single output.

The following figure shows a block having input X(s), output Y(s) and the transfer function G(s).

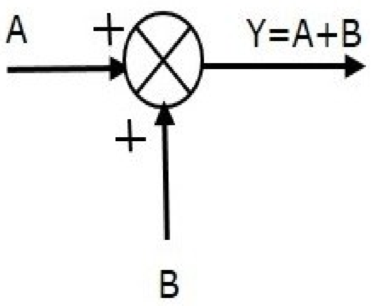
Transfer Function, $$G(s) = \frac{Y(s)}{X(s)}$$

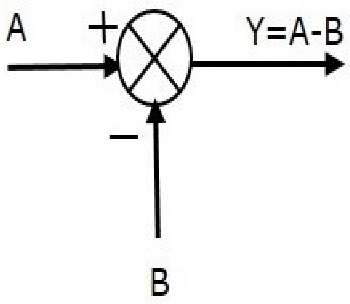
$$\Rightarrow Y(s) = G(s)X(s)$$

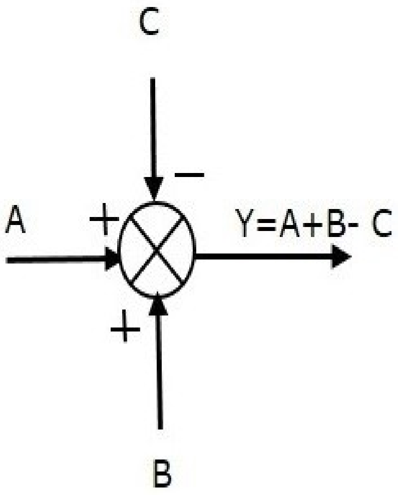
Summing Point

The summing point is represented with a circle having cross (X) inside it. It has two or more inputs and single output. It produces the algebraic sum of the inputs. It also performs the summation or subtraction or combination of summation and subtraction of the inputs based on the polarity of the inputs. Let us see these three operations one by one.

The following figure shows the summing point with two inputs (A, B) and one output (Y). Here, the inputs A and B have a positive sign. So, the summing point produces the output, Y as **sum of A and B.**



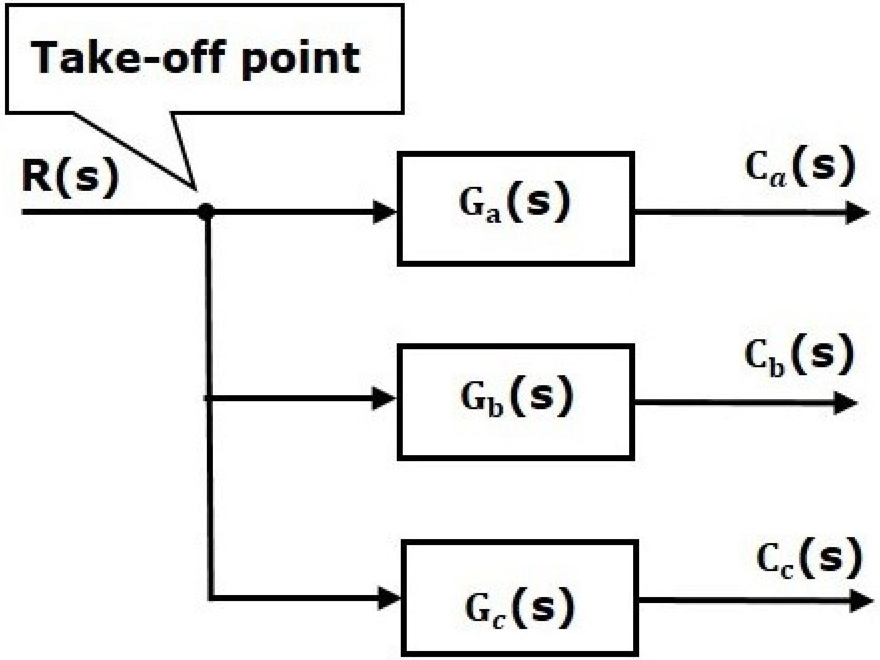


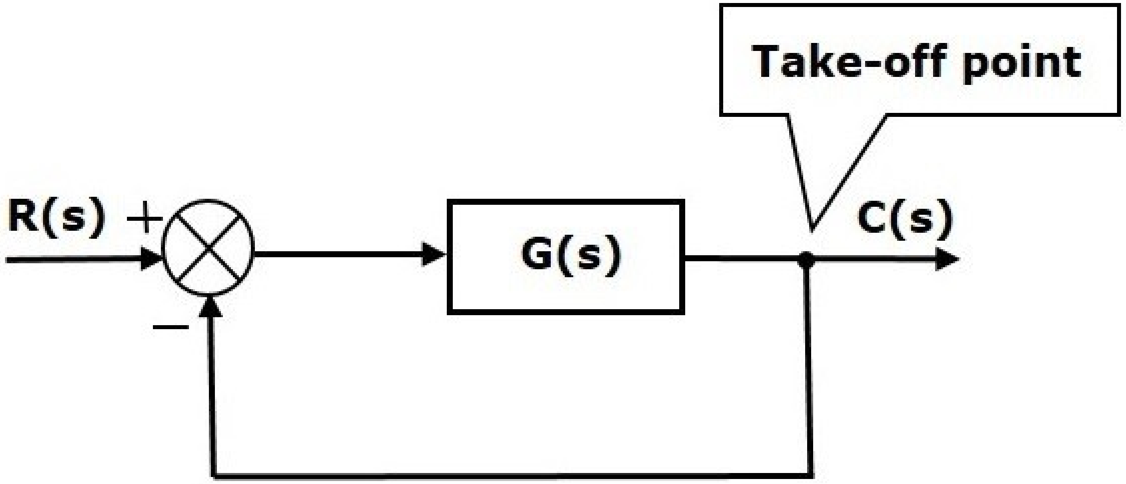


Take-off Point

The take-off point is a point from which the same input signal can be passed through more than one branch. That means with the help of take-off point, we can apply the same input to one or more blocks, summing points.

In the following figure, the take-off point is used to connect the same input, R(s) to two more blocks.

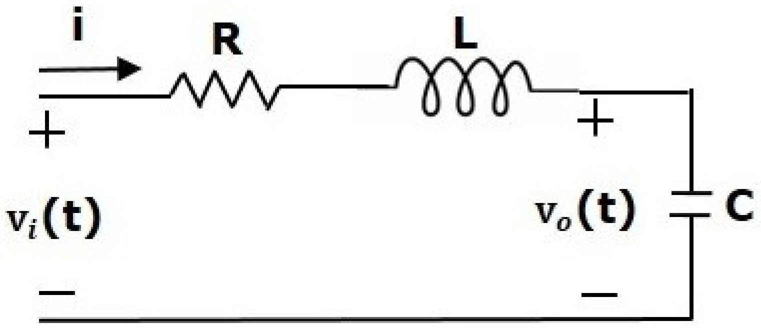




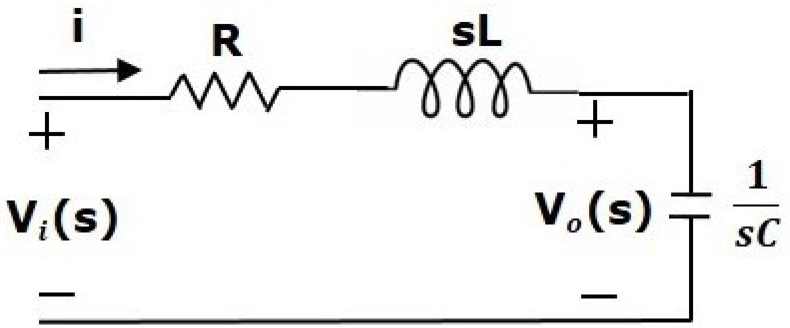
Block Diagram Representation of Electrical Systems

In this section, let us represent an electrical system with a block diagram. Electrical systems contain mainly three basic elements which are **resistor, inductor and capacitor**.

Consider a series of RLC circuit as shown in the following figure. Where, $V_i(t)$ and $V_o(t)$ are the input and output voltages. Let i(t) be the current passing through the circuit. This circuit is in time domain.



By applying the Laplace transform to this circuit, will get the circuit in s-domain. The circuit is as shown in the following figure.



From the above we can write

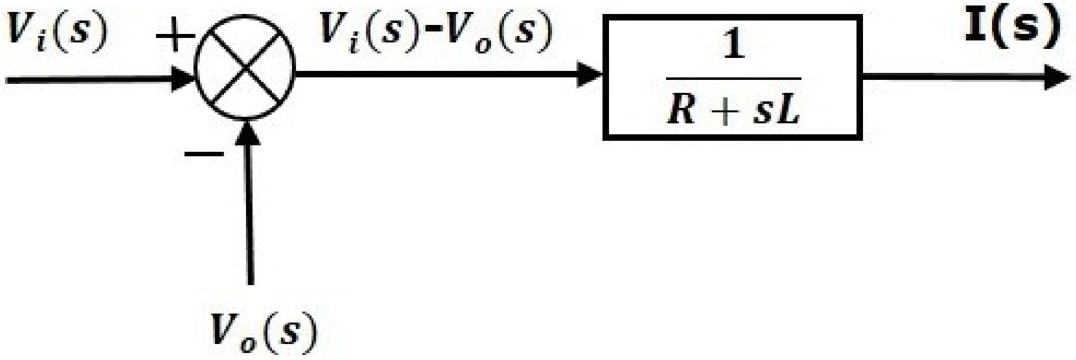
$$I(s) = \frac{V_i(s) - V_o(s)}{R + sL}$$

$$\Rightarrow I(s) = \left\{\frac{1}{R+sL}\right\}\{V_i(s) - V_o(s)\} \qquad \textbf{(Equation 1)}$$

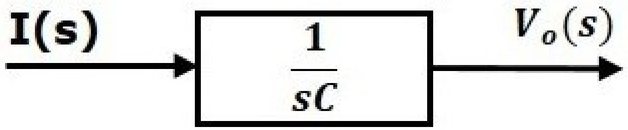
$$V_o(s) = \left(\frac{1}{sC}\right) I(s) \qquad \textbf{(Equation 2)}$$

Let us now draw the block diagrams for these two equations individually. And then combine those block diagrams properly in order to get the overall block diagram of series of RLC Circuit (s-domain).

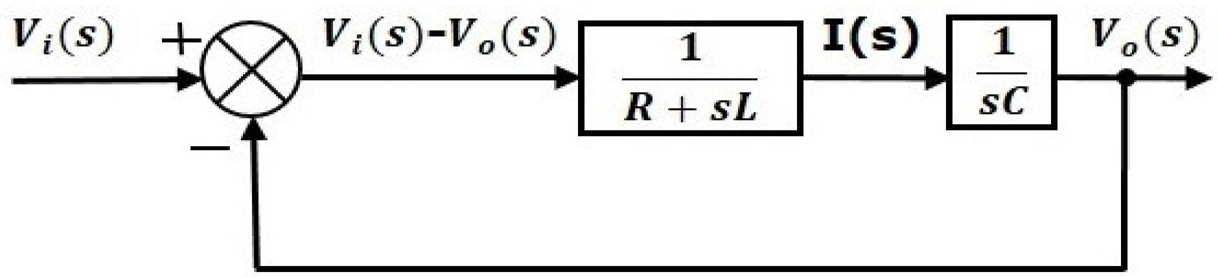
Equation 1 can be implemented with a block having the transfer function, $\frac{1}{R+sL}$. The input and output of this block are $\{V_i(s)-V_o(s)\}$ and I_s. We require a summing point to get $\{V_i(s)-V_o(s)\}$. The block diagram of Equation 1 is shown in the following figure.



Equation 2 can be implemented with a block having transfer function, $\frac{1}{sC}$. The input and output are I_s and $V_o(s)$. The block diagram of Equation 2 is shown in the following figure.



The overall block diagram of the series of RLC Circuit (s-domain) is shown in the following figure.



Similarly, you can draw the **block diagram** of any electrical circuit or system just by following this simple procedure.

- Convert the time domain electrical circuit into an s-domain electrical circuit by applying Laplace transform.
- Write down the equations for the current passing through all series branch elements and voltage across all shunt branches.
- Draw the block diagrams for all the above equations individually.
- Combine all these block diagrams properly in order to get the overall block diagram of the electrical circuit (s-domain).

LAPLACE TRANSFORM OF BASIC FUNCTION

Time domain function		Laplace transfer function
unit inpulse $\delta(t)$	$\rightarrow$	1
unit step $u(t)$	$\rightarrow$	$\dfrac{1}{s}$
t	$\rightarrow$	$\dfrac{1}{s^2}$
t^n	$\rightarrow$	$\dfrac{n!}{s^{n+1}}$
e^{-at}	$\rightarrow$	$\dfrac{1}{s+a}$
te^{-at}	$\rightarrow$	$\dfrac{1}{(s+a)^2}$
$t^n e^{-at}$	$\rightarrow$	$\dfrac{n!}{(s+a)^{n+1}}$
$\sin\omega_n t$	$\rightarrow$	$\dfrac{\omega_n}{s^2+\omega_n^2}$
$\cos\omega_n t$	$\rightarrow$	$\dfrac{s}{s^2+\omega_n^2}$

PROPERTIES OF LT

Item no.	Theorem	Name
1.	$\mathscr{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st}dt$	Definition
2.	$\mathscr{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathscr{L}[f_1(t)+f_2(t)] = F_1(s)+F_2(s)$	Linearity theorem
4.	$\mathscr{L}[e^{-at}f(t)] = F(s+a)$	Frequency shift theorem
5.	$\mathscr{L}[f(t-T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathscr{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathscr{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathscr{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathscr{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^{n} s^{n-k}f^{k-1}(0-)$	Differentiation theorem
10.	$\mathscr{L}\left[\int_{0-}^{t} f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s\to 0} sF(s)$	Final value theorem[1]
12.	$f(0+) = \lim_{s\to\infty} sF(s)$	Initial value theorem[2]

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Lecture 3

BLOCK DIAGRAM ALGEBRA

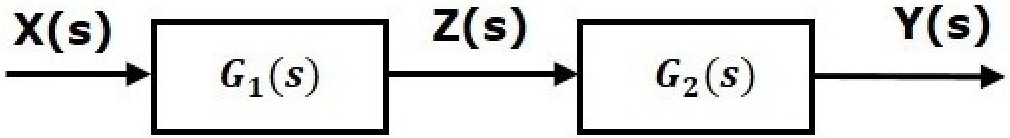
Basic Connections for Blocks

- There are three basic types of connections between two blocks.

1. Series Connection.
2. Parallel Connection.
3. Feedback Connection.

Series Connection:

Series connection is also called **cascade connection**. In the following figure, two blocks having transfer functions $G_1(s)$ and $G_2(s)$ are connected in series



For this combination, we will get the output Y(s) as

$$Y(s) = G_2(s)Z(s)$$

$$\text{Where, } Z(s) = G_1(s)X(s)$$

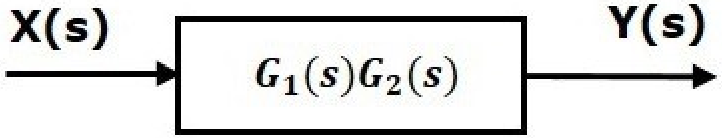
$$\text{So, } Y(s) = G_2(s)[G_1(s)X(s)] = G_1(s)\, G_2(s)X(s)$$

Compare this equation with the standard form of the output equation,

$$Y(s) = G(s)X(s)$$

$$\text{Where, } G(s) = G_1(s)\, G_2(s)$$

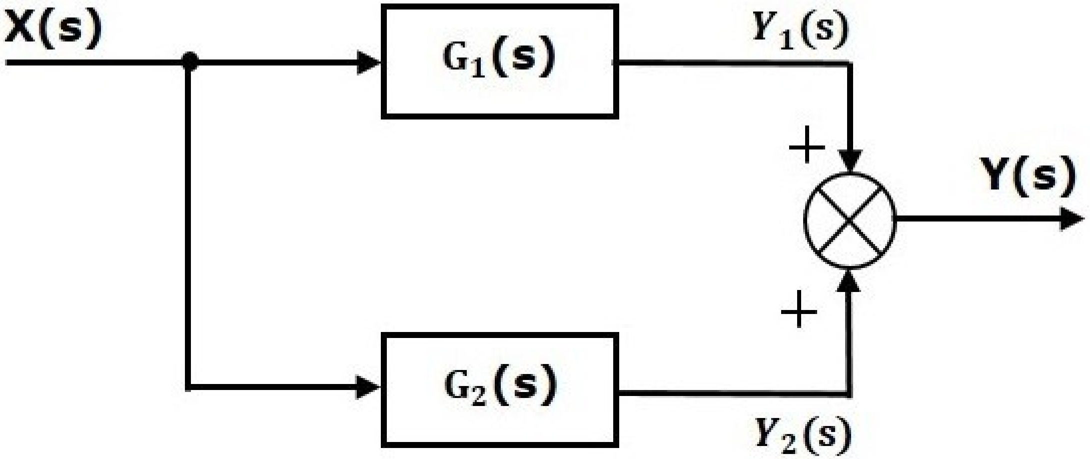
That means we can represent the **series connection** of two blocks with a single block. The transfer function of this single block is the **product of the transfer functions** of those two blocks. The equivalent block diagram is shown below.



Parallel Connection

The blocks which are connected in **parallel** will have the **same input**. In the following figure, two blocks having transfer functions $G_1(s)$ and $G_2(s)$ are connected in parallel. The outputs of these two blocks are connected to the summing point.

The blocks which are connected in **parallel** will have the **same input**. In the following figure, two blocks having transfer functions (s) and (s) are connected in parallel. The outputs of these two blocks are connected to the summing point.



For this combination, we will get the output Y(s) as

$$Y(s) = Y_1(s) + Y_2(s)$$

Where, $Y_1(s) = G_1(s)X(s)$ and $Y_2(s) = G_2(s)X(s)$

For this combination, we will get the output Y(s) as
Y(s) = (s) + (s)
Where, (s) = (s)X(s) and (s) = (s)X(s)

$$Y(s) = G_1(s)X(s) + G_2(s)X(s)$$

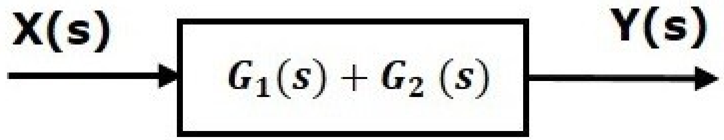
$$\text{So, } Y(s) = \{G_1(s) + G_2(s)\}\ X(s)$$

Compare this equation with the standard form of the output equation,

$$Y(s) = G(s)X(s)$$

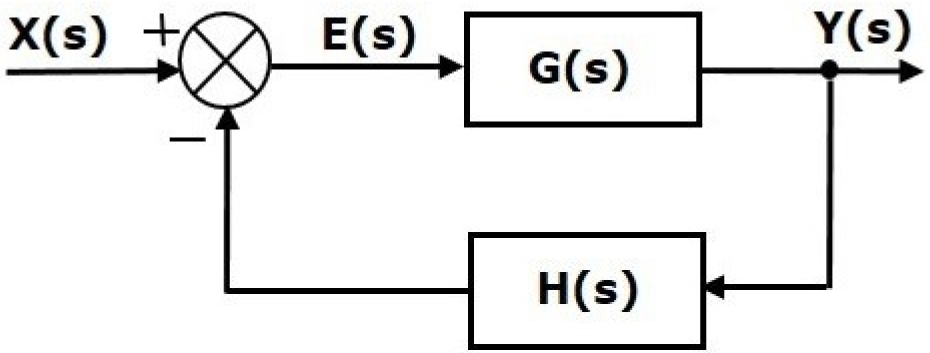
$$\text{Where, } G(s) = G_1(s) + G_2(s)$$

That means we can represent the **parallel connection** of two blocks with a single block. The transfer function of this single block is the **sum of the transfer functions** of those two blocks. The equivalent block diagram is shown below.



Feedback Connection

As we discussed earlier, there are two types of **feedback** — positive feedback and negative feedback. The following figure shows negative feedback control system. Here, two blocks having transfer functions G(s) and H(s) form a closed loop.



The output of the summing point is $E(s) = X(s) - H(s)Y(s)$

The output Y(s) is $Y(s) = E(s)\,G(s)$

Now substituting E(s) value in above equation,

$$Y(s) = \{X(s) - H(s)Y(s)\}\ G(s)$$

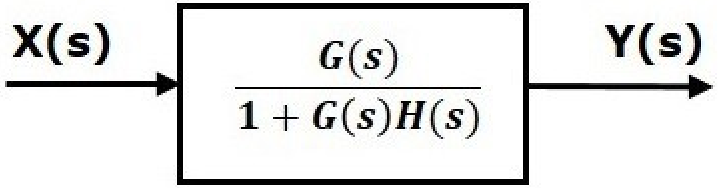
$$Y(s)\ \{1+G(s)H(s)\} = X(s)\ G(S)$$

$$\text{so, } \frac{Y(s)}{X(s)} = \frac{G(s)}{1+G(s)H(s)}$$

Therefore, the negative feedback closed loop transfer function is

$$\frac{G(s)}{1+G(s)H(s)}$$

This means we can represent the negative feedback connection of two blocks with a single block. The transfer function of this single block is the closed loop transfer function of the negative feedback. The equivalent block diagram is shown below.



Similarly, you can represent the positive feedback connection of two blocks with a single block. The transfer function of this single block is the closed loop transfer function of the positive feedback,

Similarly, you can represent the positive feedback connection of two blocks with a single block. The transfer function of this single block is the closed loop transfer function of the positive feedback,

$$\frac{G(s)}{1 - G(s)H(s)}$$

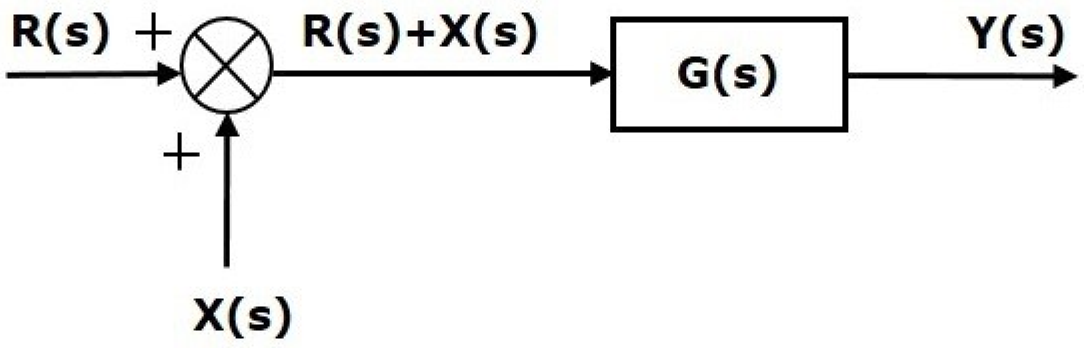
Block Diagram Algebra for Summing Points

There are two possibilities of shifting summing points with respect to blocks −

1. Shifting summing point after the block
2. Shifting summing point before the block

Shifting Summing Point After the Block

Consider the block diagram shown in the following figure. Here, the summing point is present before the block.



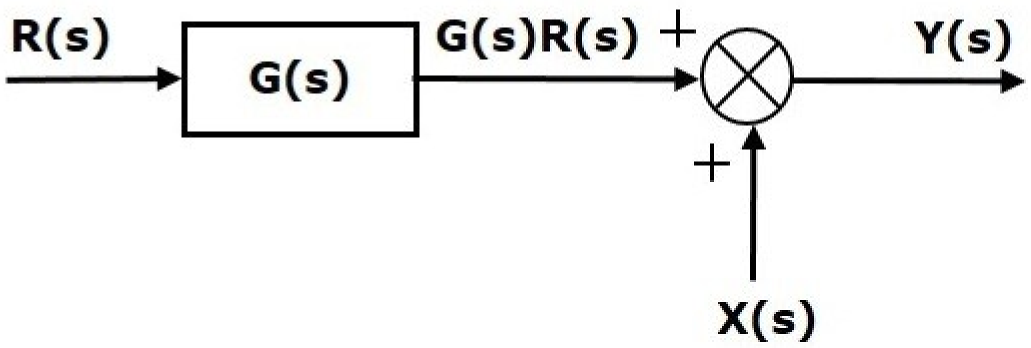
Summing point has two inputs R(s) and X(s). The output of it is {R(s) + X(s)}

So, the input to the block G(s) is {R(s) + X(s)} and the output of it is

$$Y(s) = G(s)\{R(s) + X(s)\}$$

so, $Y(s) = G(s)R(s) + G(s)X(s)$ ……..**Equation 1**

Now, shift the summing point after the block. This block diagram is shown in the following figure.

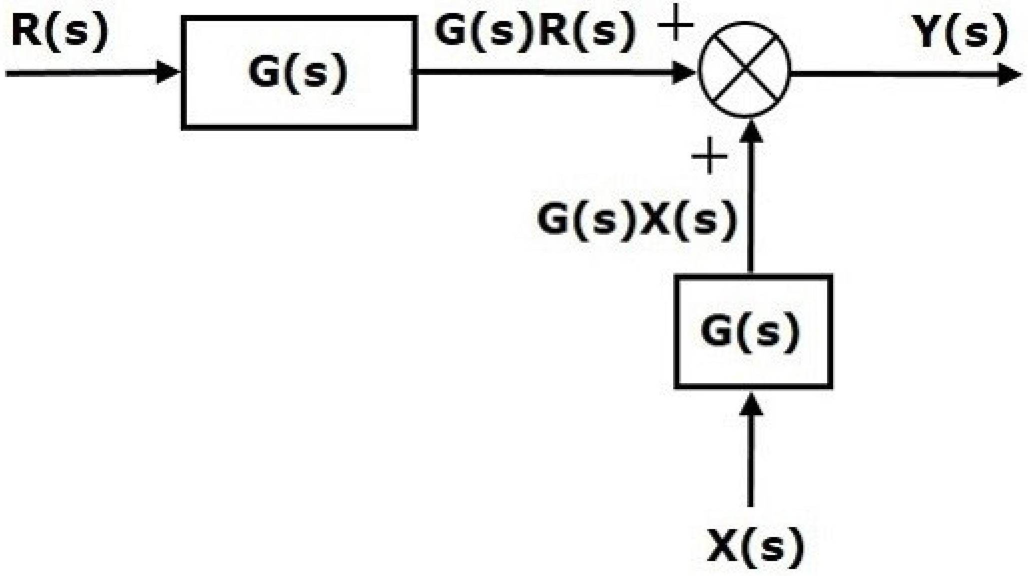


Output of the block G(s) is G(s)R(s)

The output of summing point is $Y(s) = G(s) R(s) + X(s)$…...**Equation 2**

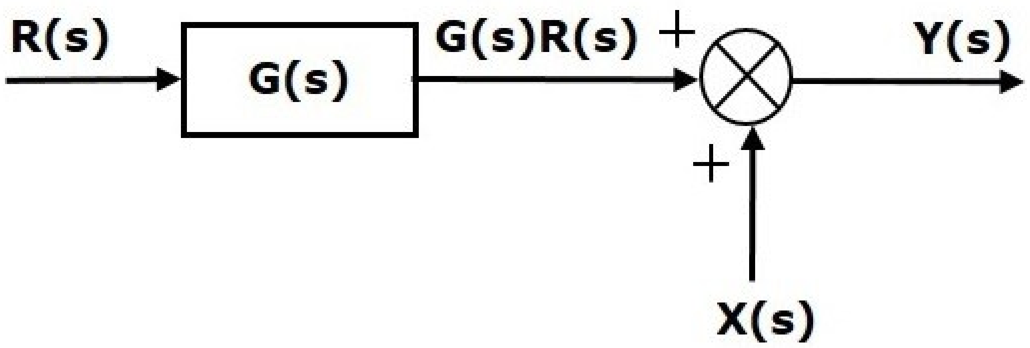
Compare Equation 1 and Equation 2.

The first term G(S) R(S) is same in both the equations. But, there is difference in the second term. In order to get the second term also same, we require one more block G(s). It is having input X(s) and the output of this block is given as input to summing point instead of X(s). This block diagram is showed in following figure



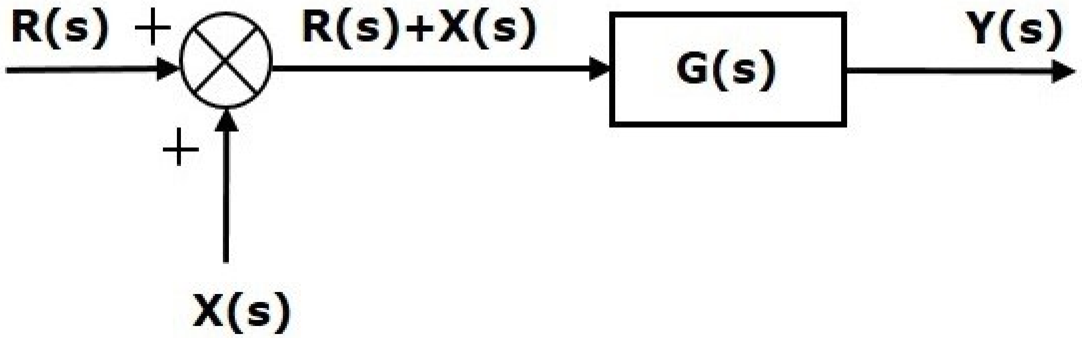
Shifting Summing Point Before the Block

Consider the block diagram shown in the following figure. Here, the summing point is present after the block.



Output of this block diagram is $Y(s) = G(s)\,R(s) + X(s)$……..**Equation 1**

Now, shift the summing point before the block. This block diagram is shown in the following figure.



Output of this block diagram is $Y(s) = G(s) R(s) + G(s)X(s)$......**Equation 2**

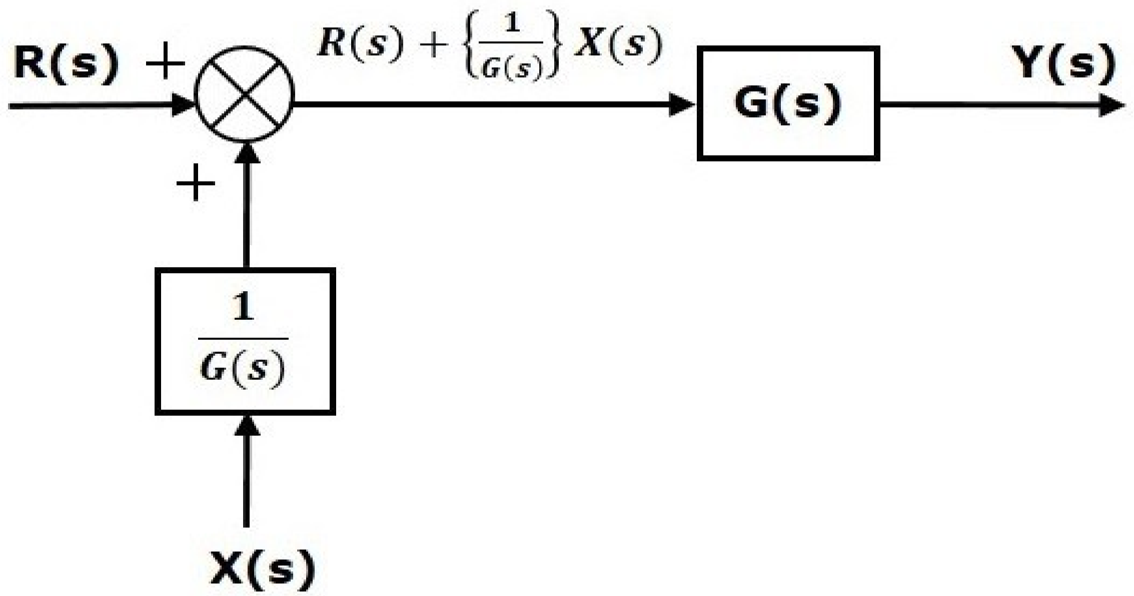
The first term $G(s) R(s)$ is same in both equations. But, there is difference in the second term. In order to get the second term also same, we require one more block $\frac{1}{G(s)}$. It is having the input $X(s)$ and the output of this block is given as input to summing point instead of $X(S)$.

This block diagram is shown in the following figure.

Output of this block diagram is $Y(s) = G(s) R(s) + G(s)X(s)$......**Equation 2**

The first term $G(s) R(s)$ is same in both equations. But, there is difference in the second term. In order to get the second term also same, we require one more block It is having the input $X(s)$ and the output of this block is given as input to summing point instead of $X(S)$.

This block diagram is shown in the following figure.



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Lecture 4

Block Diagram Algebra for Take-off Points

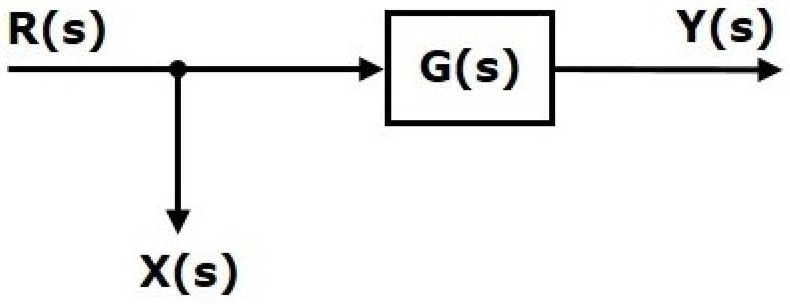
There are two possibilities of shifting the take-off points with respect to blocks −

1. Shifting take-off point after the block
2. Shifting take-off point before the block

Let us now see what kind of arrangements are to be done in the above two cases, one by one.

Shifting Take-off Point After the Block

Consider the block diagram shown in the following figure. In this case, the take-off point is present before the block.

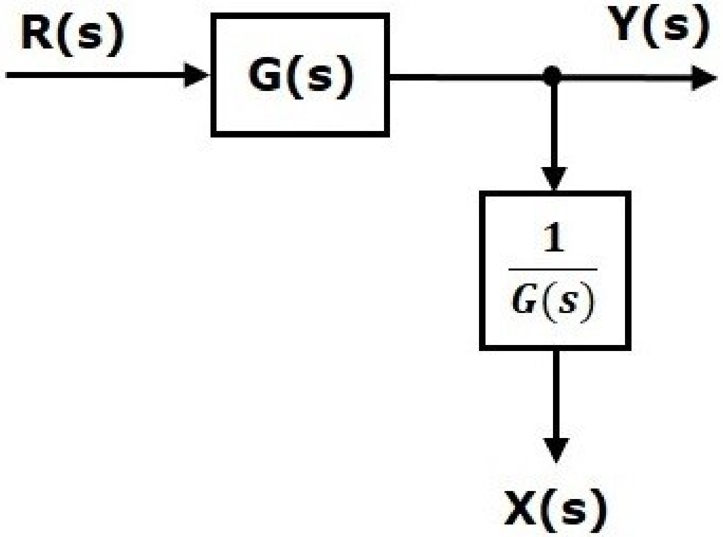


Here, $X(s) = R(s)$ and $Y(s) = G(s)\,R(s)$

When you shift the take-off point after the block, the output $Y(s)$ will be same. But, there is difference in $X(s)$ value. So, in order to get the same $X(s)$ value, we require one more block. It is having $Y(s)$ and the output is $X(s)$. The block diagram is showed in the figure.

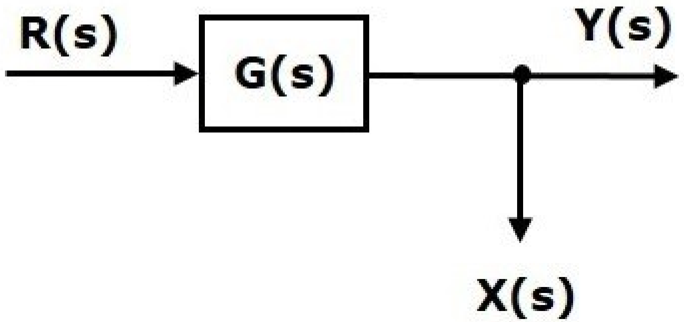
Here, $X(s) = R(s)$ and $Y(s) = G(s)\,R(s)$

When you shift the take-off point after the block, the output $Y(s)$ will be same. But, there is difference in $X(s)$ value. So, in order to get the same $X(s)$ value, we require one more block $\frac{1}{G(s)}$. It is having $Y(s)$ and the output is $X(s)$. The block diagram is showed in the figure.



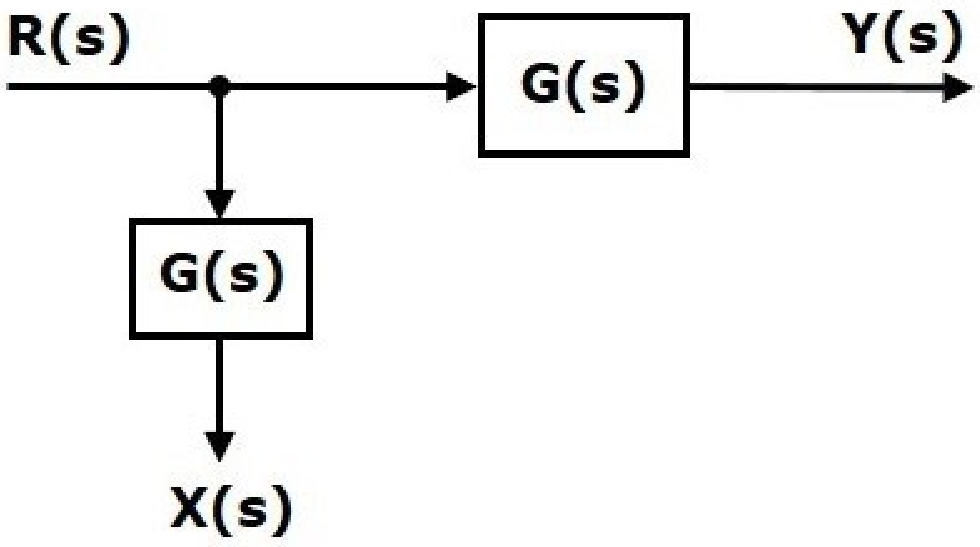
Shifting Take-off Point Before the Block

Consider the block diagram shown in the following figure. Here, the take-off point is present after the block.



Here, X(s) = Y(s) = G(s) R(s)

When you shift the take-off point before the block, the output Y(s) will be same. But, there is difference in X(s) value. So, in order to get the same X(s) value, we require one more block G(s) It is having R(s) and the output is X(s). The block diagram is showed in the figure.


R(s)
G(s)
Y(s)
G(s)
X(s)

BLOCK DIAGRAM REDUCTION

Block Diagram Reduction Rules

Follow these rules for simplifying (reducing) the block diagram, which is having many blocks, summing points and take-off points.

Rule 1 − Check for the blocks connected in series and simplify.

Rule 2 − Check for the blocks connected in parallel and simplify.

Rule 3 − Check for the blocks connected in feedback loop and simplify.

Rule 4 − If there is difficulty with take-off point while simplifying, shift it towards right.

Rule 5 − If there is difficulty with summing point while simplifying, shift it towards left.

Rule 6 − Repeat the above steps till you get the simplified form, i.e., single block.

Note − The transfer function present in this single block is the transfer function of the overall block diagram.

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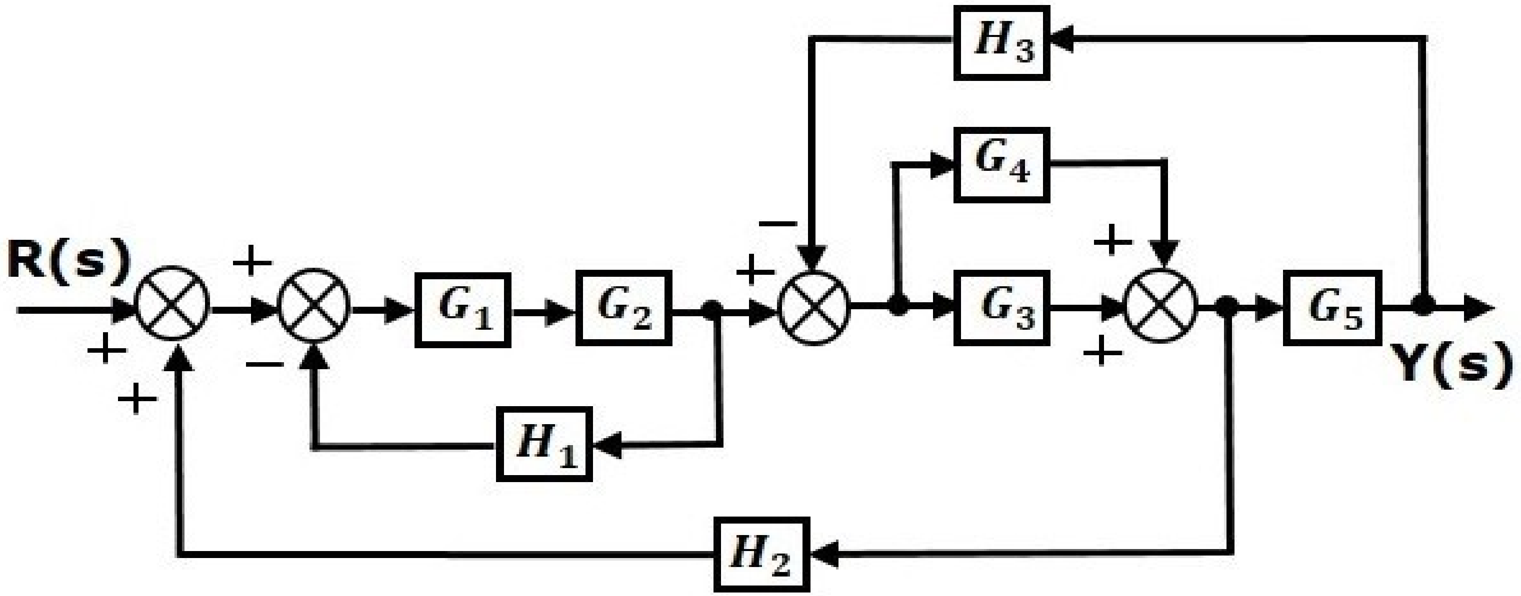
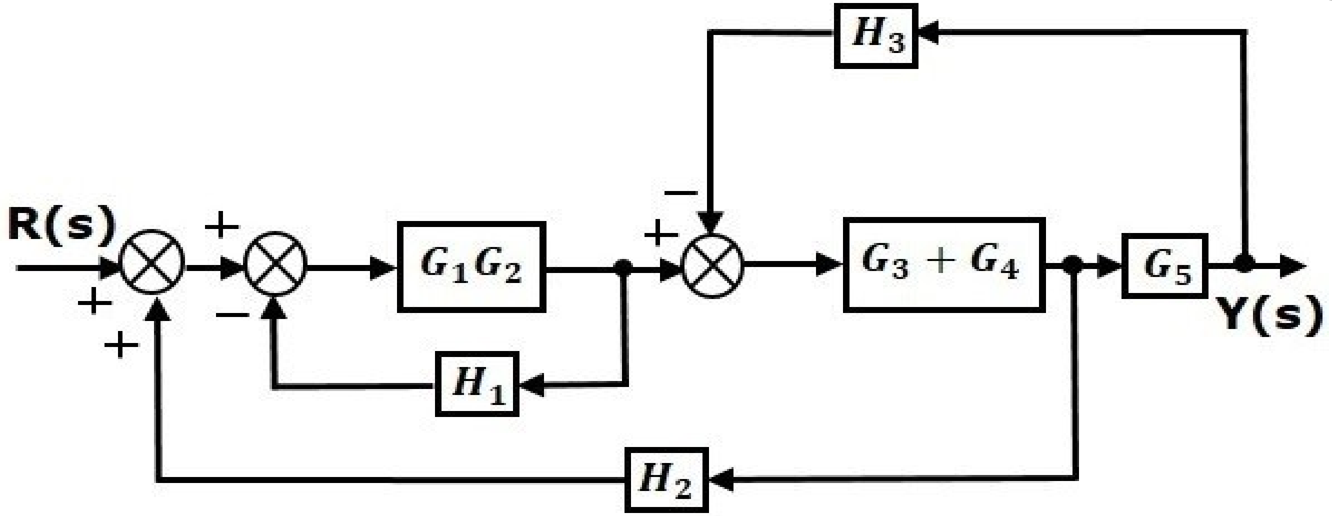
Example:

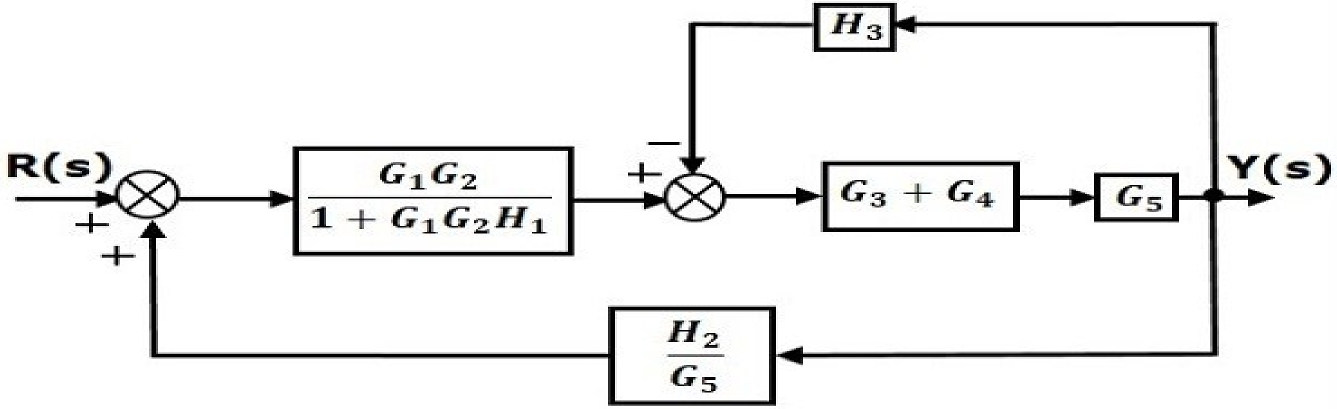
Consider the block diagram shown in the following figure. Let us simplify (reduce) this block diagram using the block diagram reduction rules.



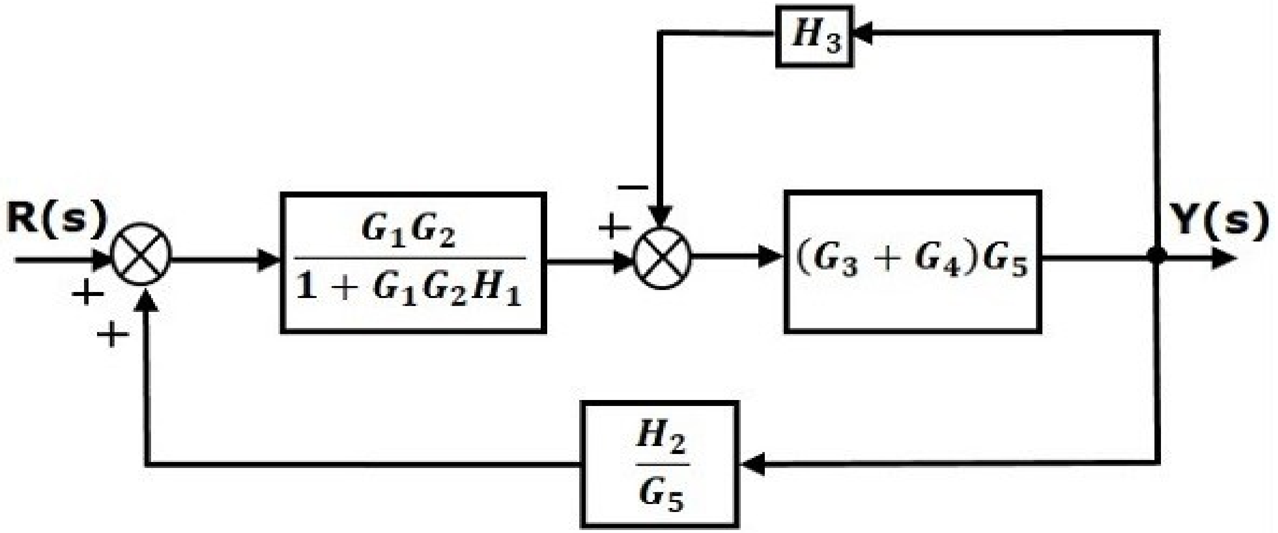
Step 1 – Use Rule 1 for blocks G_1 and G_2. Use Rule 2 for blocks G_3 and G_4. The modified block diagram is shown in the following figure.



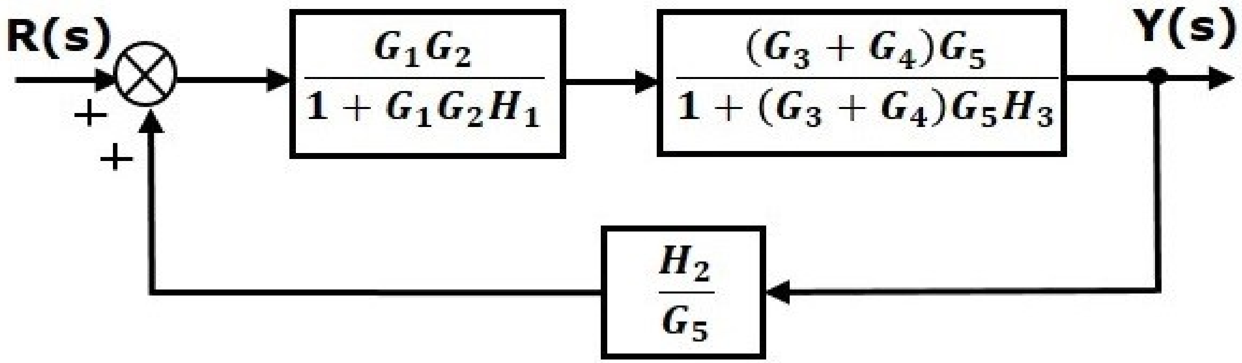
Step 2 – Use Rule 3 for blocks G_1G_2 and H_1. Use Rule 4 for shifting take-off point after the block G_5. The modified block diagram is shown in the following figure.



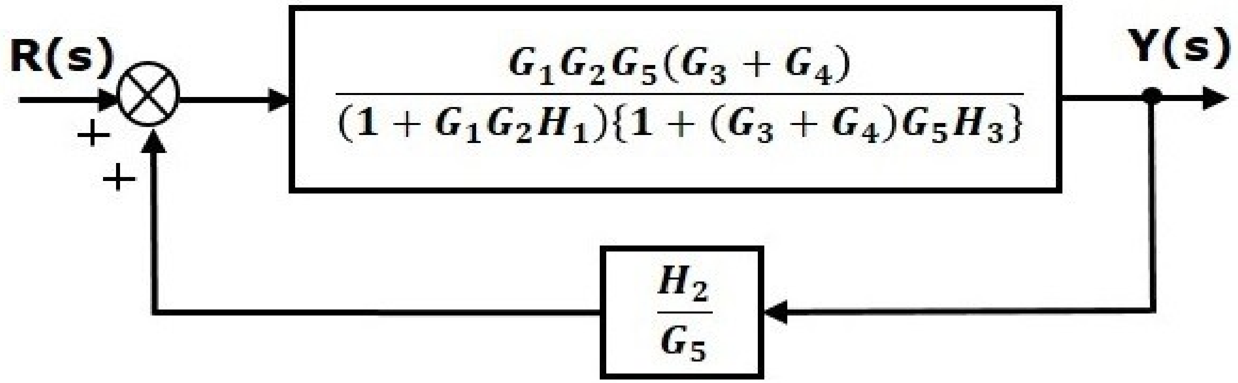
Step 3 – Use Rule 1 for blocks $(G_3 + G_4)$ and G_5. The modified block diagram is shown in the following figure.



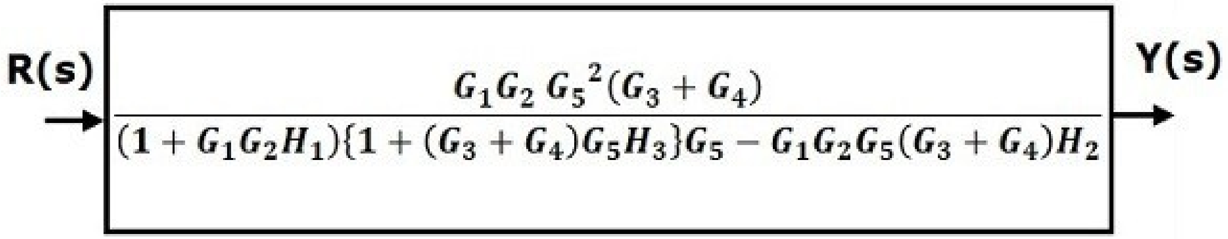
Step 4 – Use Rule 3 for blocks $(G_3 + G_4)G_5$ and H_3. The modified block diagram is shown in the following figure.



Step 5 − Use Rule 1 for blocks connected in series. The modified block diagram is shown in the following figure.



Step 6 − Use Rule 3 for blocks connected in feedback loop. The modified block diagram is shown in the following figure. This is the simplified block diagram.



Therefore, the transfer function of the system is

$$\frac{Y(s)}{R(s)} = \frac{G_1 G_2 G_5^2 (G_3 + G_4)}{(1 + G_1 G_2 H_1)\{1 + (G_3 + G_4) G_5 H_3\} G_5 - G_1 G_2 G_5 (G_3 + G_4) H_2}$$

Note − Follow these steps in order to calculate the transfer function of the block diagram having multiple inputs.

Step 1 − Find the transfer function of block diagram by considering one input at a time and make the remaining inputs as zero.

Step 2 − Repeat step 1 for remaining inputs.

Step 3 − Get the overall transfer function by adding all those transfer functions.

The block diagram reduction process takes more time for complicated systems. Because, we have to draw the (partially simplified) block diagram after each step. So, to overcome this drawback, use signal flow graphs (representation).

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Lecture 5

SIGNAL FLOW GRAPH

Signal flow graph is a graphical representation of algebraic equations. In this lecture, let us discuss the basic concepts related signal flow graph and also learn how to draw signal flow graphs.

Basic Elements of Signal Flow Graph

Nodes and branches are the basic elements of signal flow graph.

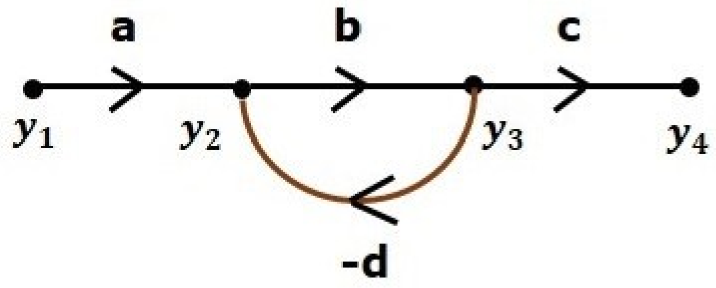
Node:

Node is a point which represents either a variable or a signal. There are three types of nodes — input node, output node and mixed node.

1. **Input Node** − It is a node, which has only outgoing branches.
2. **Output Node** − It is a node, which has only incoming branches.
3. **Mixed Node** − It is a node, which has both incoming and outgoing branches.

Example

Let us consider the following signal flow graph to identify these nodes.



- The **nodes** present in this signal flow graph are $\mathbf{y_1}$**,** $\mathbf{y_2}$**,** $\mathbf{y_3}$ and $\mathbf{y_4}$.
- $\mathbf{y_1}$ and $\mathbf{y_4}$ are the **input node** and **output node** respectively.
- $\mathbf{y_2}$ and $\mathbf{y_3}$ are **mixed nodes**.

Branch

Branch is a line segment which joins two nodes. It has both **gain** and **direction**. For example, there are four branches in the above signal flow graph. These branches have **gains** of **a, b, c** and **-d**.

Construction of Signal Flow Graph

Let us construct a signal flow graph by considering the following algebraic equations −

$$y_2 = a_{12}y_1 + a_{42}y_4$$

$$y_3 = a_{23}y_2 + a_{53}y_5$$

$$y_4 = a_{34}y_3$$

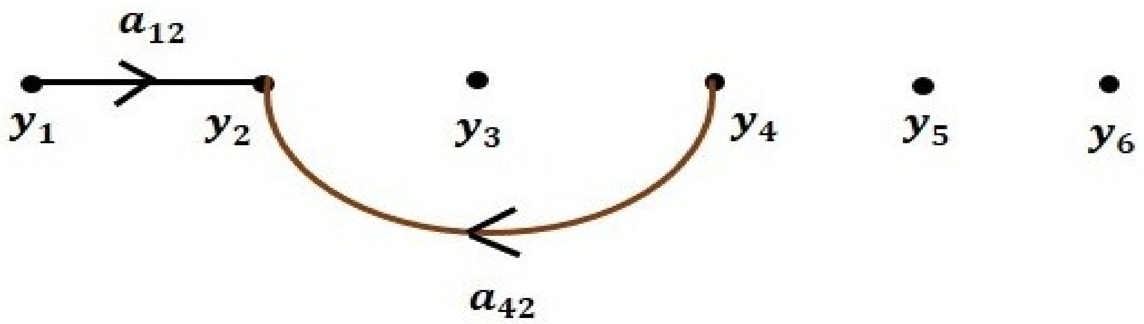
$$y_5 = a_{45}y_4 + a_{35}y_3$$

$$y_6 = a_{56}y_5$$

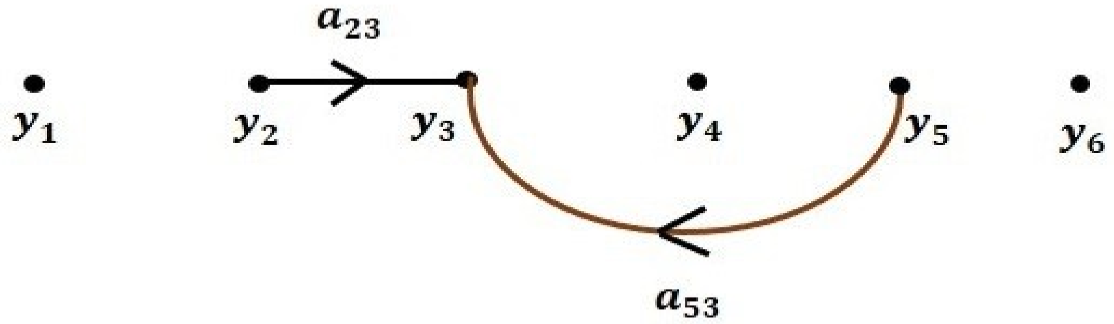
There will be six **nodes** (y_1, y_2, y_3, y_4, y_5 and y_6) and eight **branches** in this signal flow graph. The gains of the branches are a_{12}, a_{23}, a_{34}, a_{45}, a_{56}, a_{42}, a_{53} and a_{35}.

To get the overall signal flow graph, draw the signal flow graph for each equation, then combine all these signal flow graphs and then follow the steps given below –

Step 1 − Signal flow graph for $y_2 = a_{12}y_1 + a_{42}\,y_4$ is shown in the following figure.

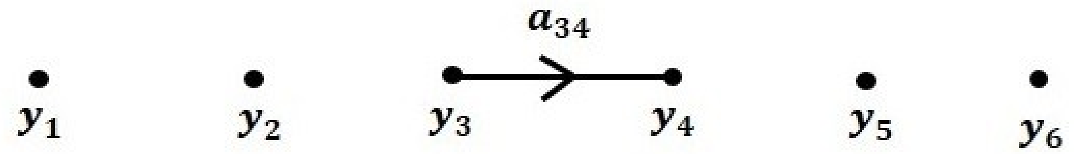


Step 2 − Signal flow graph for $y_3 = a_{23}y_2 + a_{53}\,y_5$ is shown in the following figure.



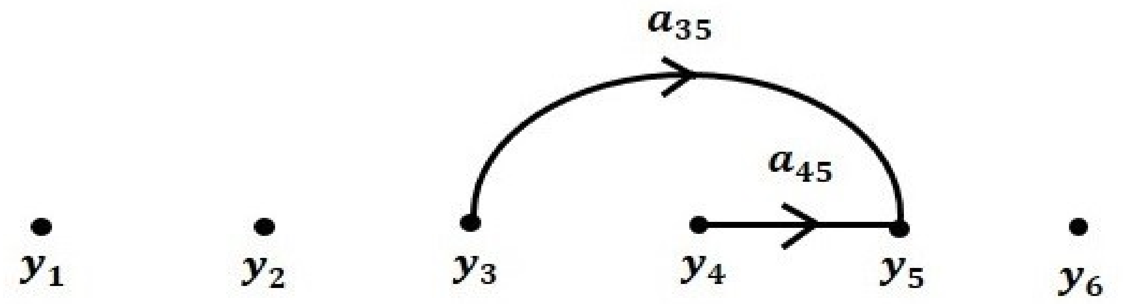
Step 3 − Signal flow graph for = is shown in the following figure.

Step 3 − Signal flow graph for $y_4 = a_{34} y_3$ is shown in the following figure.



Step 4 − Signal flow graph for = is shown in the following figure.

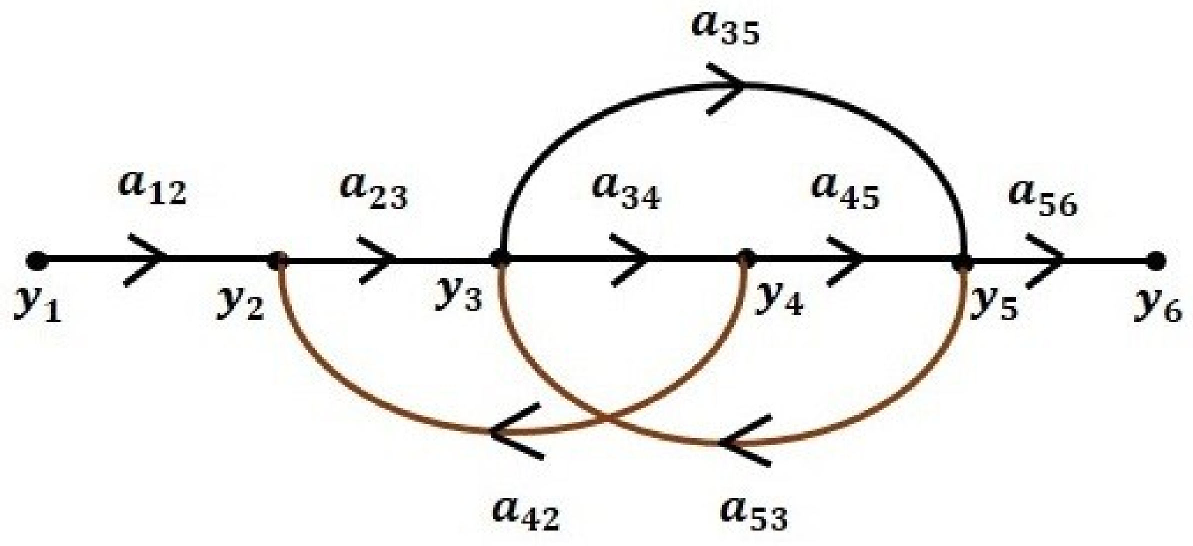
Step 4 − Signal flow graph for $y_5 = a_{45} y_4 + a_{35}\, y_3$ is shown in the following figure.



Step 5 − Signal fl igure.

Step 5 − Signal flow graph for $y_6 = a_{56} y_5$ is shown in the following figure.

Step 6 − Signal flow graph of overall system is shown in the following figure.



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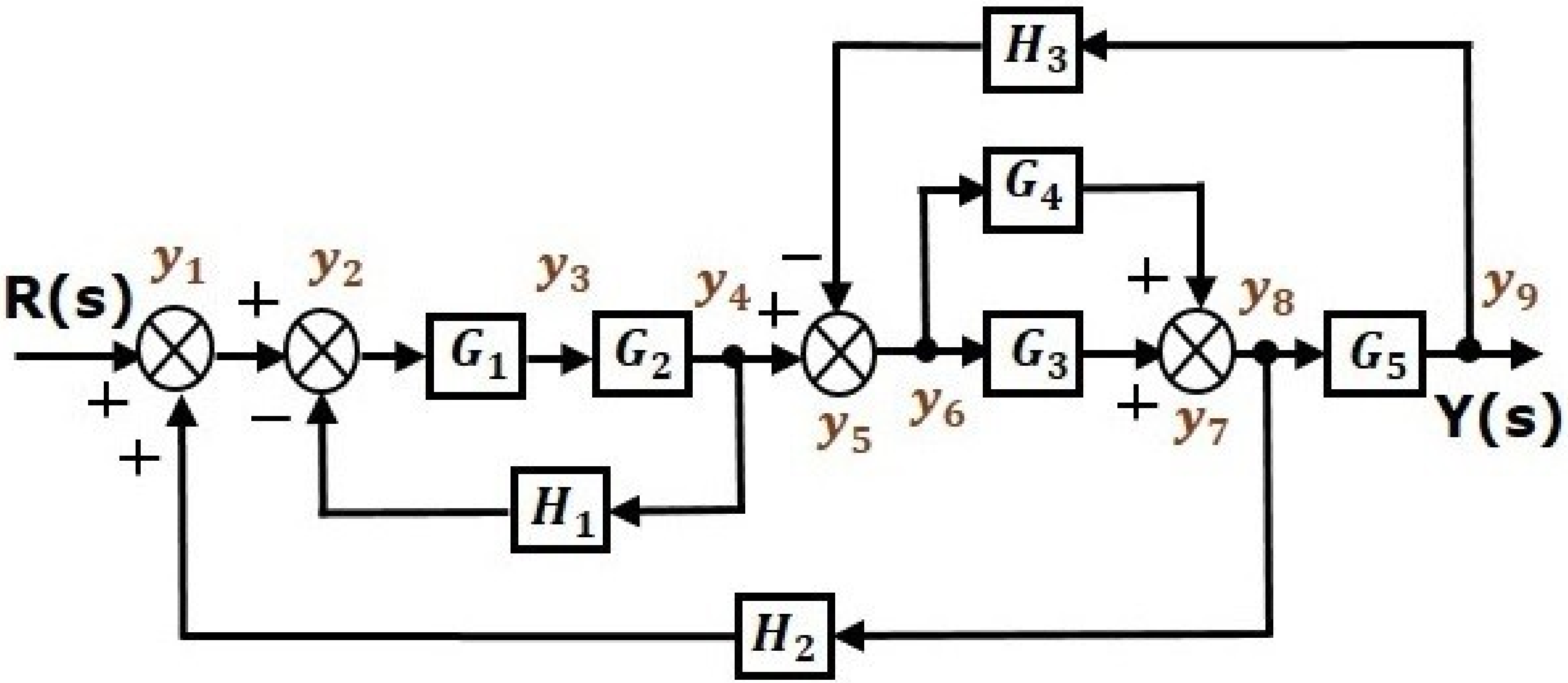
Conversion of Block Diagrams into Signal Flow Graphs

Follow these steps for converting a block diagram into its equivalent signal flow graph.

- Represent all the signals, variables, summing points and take-off points of block diagram as **nodes** in signal flow graph.
- Represent the blocks of block diagram as **branches** in signal flow graph.
- Represent the transfer functions inside the blocks of block diagram as **gains** of the branches in signal flow graph.
- Connect the nodes as per the block diagram. If there is connection between two nodes (but there is no block in between), then represent the gain of the branch as one. **For example**, between summing points, between summing point and takeoff point, between input and summing point, between take-off point and output.

Example 1:

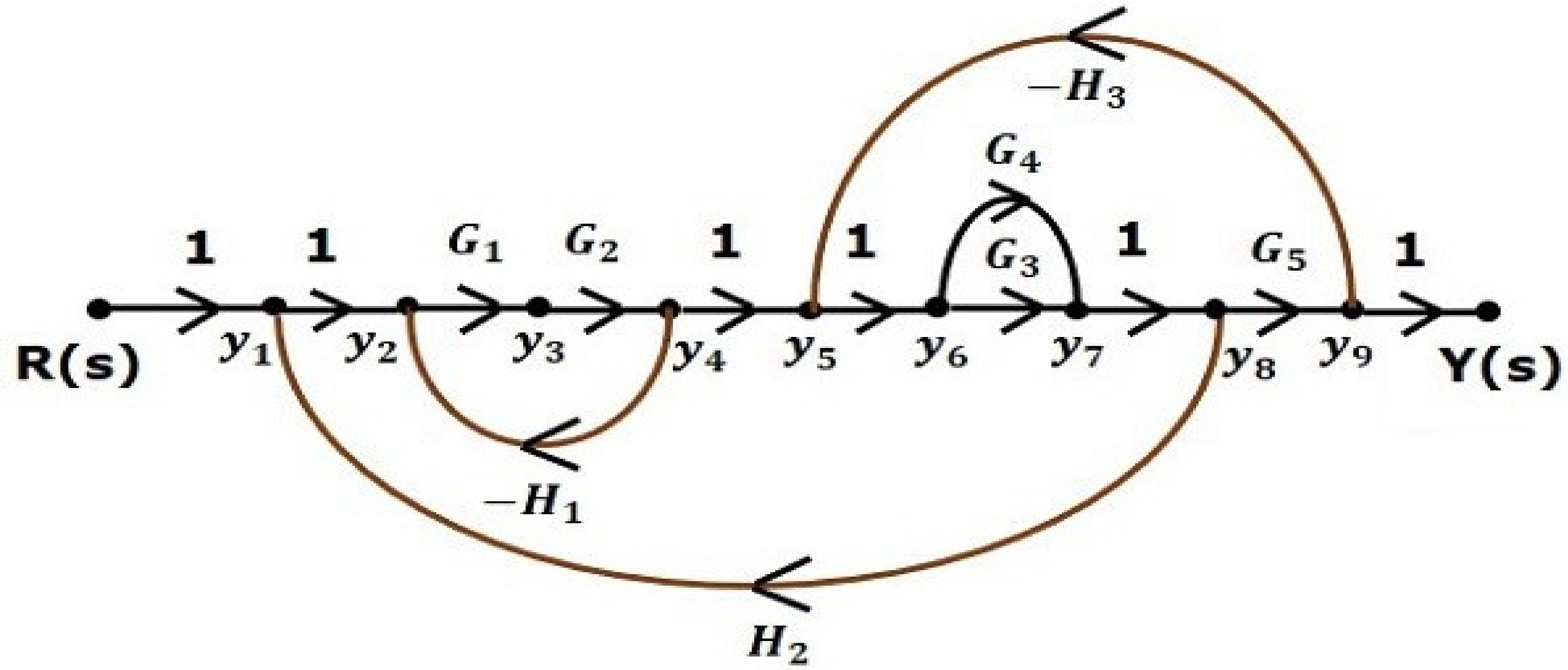
Let us convert the following block diagram into its equivalent signal flow graph.



Represent the input signal R(s) and output signal C(s) of block diagram as input node R(s) and output node C(s) of signal flow graph.

Just for reference, the remaining nodes (y_1 to y_9) are labelled in the block diagram. There are nine nodes other than input and output nodes. That is four nodes for four summing points, four nodes for four take-off points and one node for the variable between blocks G_1 and G_2.

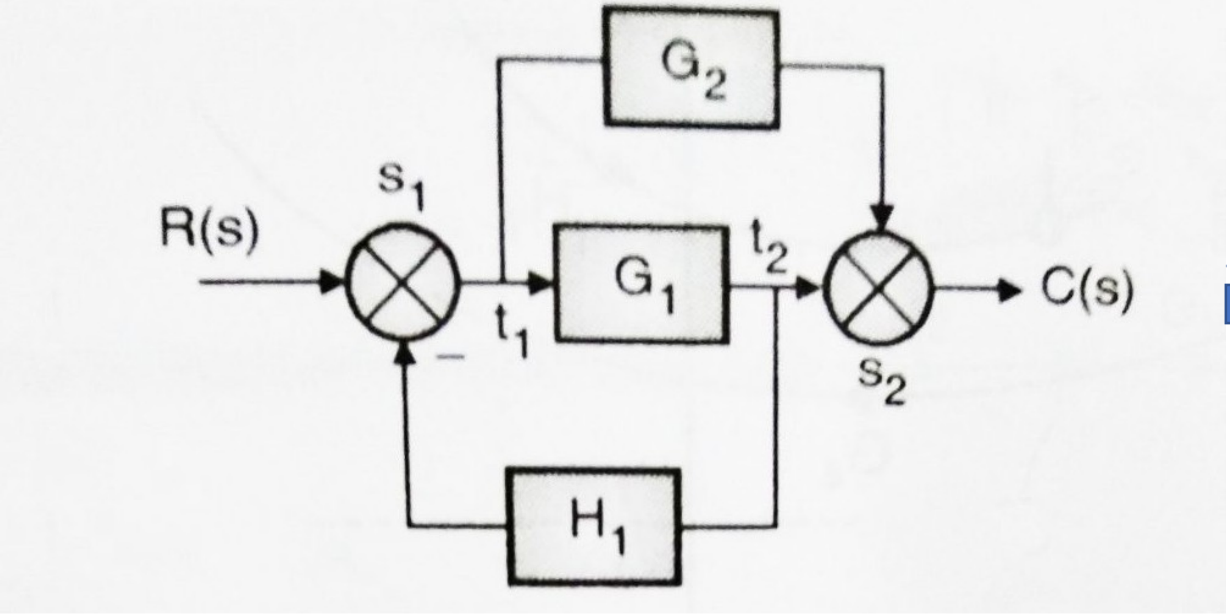
The following figure shows the equivalent signal flow graph.

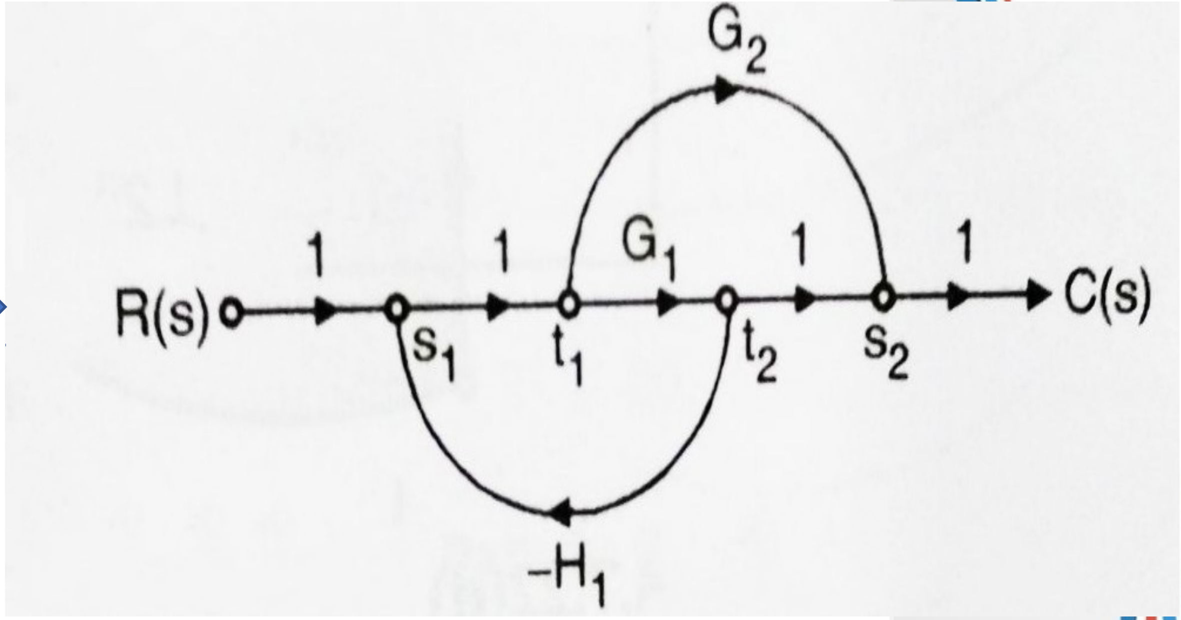


With the help of Mason's gain formula (discussed in the next Lecture), you can calculate the transfer function of this signal flow graph. This is the advantage of signal flow graphs. Here, we no need to simplify (reduce) the signal flow graphs for calculating the transfer function.

Example 2:

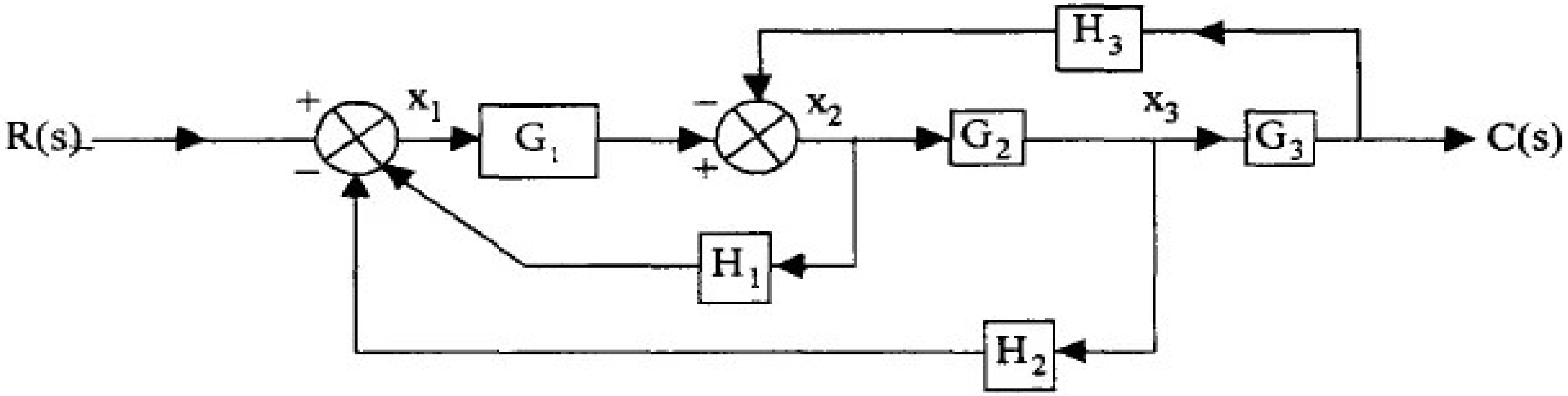
Let us convert the following block diagram into its equivalent signal flow graph.

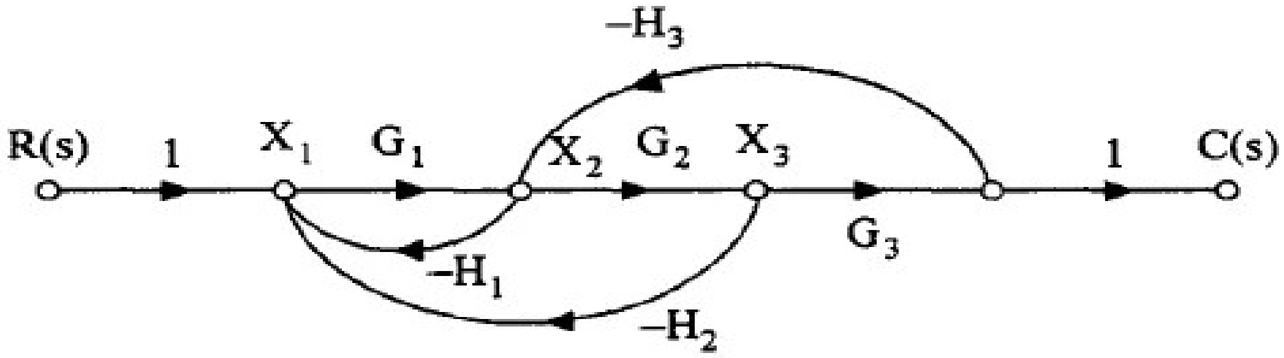




Example 3:

Let us convert the following block diagram into its equivalent signal flow graph.





Thank You